

Enrollment No: _____

PARUL UNIVERSITY
FACULTY OF ENGINEERING & TECHNOLOGY
B.Tech Mid Semester Exam

Semester: 5th
Subject Code: 303105311
Subject Name: Quant and Reasoning

Date: (14/08/2024)
Time: (1hr: 30min)
Total Marks: 40

Sr. No.	Marks
Q.1	05
(A) Answer the following Questions.	
(1) What is the average of natural numbers from 1 to 67 ?	
(2) Check whether 1440 is divisible by 15.	
(3) Find the L.C.M of 35,56 and 91.	
(4) Find two numbers such that their mean proportional is 6 and third proportional is 20.25 .	
(5) The sum of ages of 5 children born at the intervals of 3 years each is 50 years. What is the age of the youngest child?	
(B) Fill in the blanks	
(1) The total age of A and B is 12 years more than the total age of B and C. C is _____ years younger than A .	
(2) The 17th term of AP 4, 9, 14, ...is _____ .	
(3) If 10 % of x is equal to 20 % of y , then x:y = _____ .	
(4) 81% of 2300 – 34% of 550 = _____ .	
(5) If the nth term of a GP is 128 and both the first term a and the common ratio r are 2. The number of terms in the GP is _____ .	
Q.2	12
Attempt any four.	
(1) Praveen and Naveen started a business with investments of Rs.20000 and Rs.24000 respectively. At the end of one year , the total profit they shared was Rs.1000 more than the difference in their profit shares . Find the total profit.	
(2) Find the three numbers in geometric progression having a sum of 28 and a product as 512.	
(3) 36% of the maximum marks in an examination is equal to 756 marks . what is 54% of the maximum marks in it ?	
(4) The cost of 4 bags and 3 boxes is Rs.555 and the cost of 3 bags and 4 boxes is Rs.460 . what is the cost of one bag ?	

	(5) A builder appoints three construction workers Akash, Sunil and Rakesh on one of his sites. They take 20, 30 and 60 days respectively to do a piece of work. How many days will it take Akash to complete the entire work if he is assisted by Sunil and Rakesh every third day?	
Q.3	Attempt any two questions	08
	(1) The average weight of 24 students of section A of a class is 58 Kg, whereas the average weight of 26 students of section B of the same class is 60.5 Kg. Find out average weight of all the 50 students of the class.	
	(2) It takes 20 minutes for pipe A to fill the tank completely and it takes 30 minutes for pipe B to fill the tank completely. If both the inlets are opened together, then how much time will be taken to fill the tank completely?	
	(3) Two numbers A and B are such that the sum of 5% of A and 4% of B is two-third of the sum of 6% of A and 8% of B. Find the ratio of A : B.	
Q.4	(A) What is the difference between the compound interests on Rs. 5000 for 1.5 years at 4% per annum compounded yearly and half-yearly?	05
	(B) Time taken by A to finish a piece of work is twice the time taken B and thrice the time taken by C. If all three of them work together, it takes them 2 days to complete the entire work. How much work was done by B alone ?	05
	OR	
	(B) The profit made in selling 12 m of a cloth is equal to the selling price of 4.8 m of it . Find the profit percentage made.	05